MIL-C-13351B(MR)
7 April 1971
SUPERSEDING
MIL-C-13351A(MR)
13 September 1963

MILITARY SPECIFICATION

COPPER ALLOY NUMBER 377 FORGINGS

1. SCOPE

1.1 Scope. This specification covers copper alloy number 377 (forging brass) forgings.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY
MIL-C-3993 - Copper and Copper Base Alloy Mill Products, Packaging of

STANDARDS

FEDERAL
Fed. Test Method Std. No. 151 - Metals; Test Methods
Fed. Std. No. 185 - Identification Marking of Copper and Copper Base Alloy Mill Products

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Area FORG

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

B 154 - Mercurous Nitrate Test for Copper and Copper Alloys
E 18 - Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The material shall conform to the chemical requirements specified in table I.

Table I. Chemical composition

Copper, percent	Lead percent	Iron max., percent	Zinc percent	Total other elements, max., percent
58.0-62.0	1.5-2.5	0.30	Remainder	0.50

3.1.2 Analysis shall be made regularly only for the elements specifically mentioned in table I. If, however, the presence of other elements is suspected or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.2 Hardness.

3.2.1 Forgings for time-fuze rings. The Rockwell hardness of forgings for time fuze rings at any point on the surface or section, shall be not less than B55 nor more than B70. In addition, the difference in hardness at any two points on the surface or section shall not exceed 10 Rockwell B points.

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3.2.2 Other forgings. Unless otherwise specified in the contract or order, the Rockwell hardness of forgings intended for applications other than time fuze rings shall be not less than B40.

3.3 Macrostructure. When examined as specified in 4.6.3, the forgings shall show no evidence of laminations, cracks, folds, porosity, or other injurious defects.

3.4 Mercurous nitrate test. The forgings shall withstand without cracking the mercurous nitrate test specified in 4.6.4.

3.5 Dimensional tolerances. The forgings shall conform to the dimensions and tolerances specified in the contract or order or shown on the applicable drawings.

3.6 Identification marking. When specified in the contract or order, product identification marking shall be in accordance with Fed. Std. No. 185 (see 6.2 and 6.3).

3.7 Workmanship. The forgings shall have a bright, clean surface free from oil, grease, oxidation, and scale; and shall be of uniform quality and condition; and shall be free from cracks, seams, folds, fins, cold shuts, porosity, segregations, excessive non-metallic inclusions, and other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. For forgings weighing over 1/2 pound each, the size of a lot shall be as specified in the contract or order. For forgings weighing 1/2 pound each or less, a lot shall consist of not more than 20,000 forgings of the same size, and shape.

4.3 Sampling. Samples taken for the purpose of the tests prescribed in this specification shall be selected in a manner that will represent correctly the material furnished and avoid needless destruction of finished material when samples representative of the material are available from other sources.

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4.3.1 Sampling for chemical analysis. At least one forging shall be selected from each 10,000 forgings or fraction thereof in the lot. Not less than two ounces of chips shall be obtained from each forging selected for test at such points as may be designated by either sawing, milling, or drilling. A composite sample may be made from the chips from any one forging. Each forging shall be analyzed separately. Samples for chemical analysis may be obtained from forgings used for hardness tests.

4.3.2 Sampling for tests.

4.3.2.1 Forgings weighing 1/2 pound each, or less.

4.3.2.1.1 Sampling for hardness tests. Twenty forgings shall be selected from each lot for hardness tests.

4.3.2.1.2 Sampling for macroscopic examination. Two forgings shall be selected from each lot for macroscopic examination. Samples for macroscopic examination may be selected from forgings used for the hardness test.

4.3.2.1.3 Sampling for mercurous nitrate tests. Two forgings shall be selected from each lot for the mercurous nitrate test.

4.3.2.2 Forgings over 1/2 pound. In the case of forgings exceeding 1/2 pound in weight, the number of tests shall be as specified in the contract or order.

4.3.3 Sampling for visual and dimensional examination.

4.3.3.1 Visual examination. From each lot of material a representative sample of material shall be selected in accordance with MIL-STD-105, inspection level III, acceptable quality level (AQL) 1.5 percent defective, and shall be visually examined in accordance with 4.4.1.

4.3.3.2 Dimensional examination. From each lot of material a representarive sample shall be selected in accordance with 4.3.3.1 and shall be measured in accordance with 4.4.2. The samples selected for dimensional examination may be the same as those selected for visual examination.

4.4 Examination.

4.4.1 Visual. Each sample unit selected in accordance with 4.3.3 shall be visually examined to determine compliance with the requirements for identification marking (see 3.6) and workmanship (see 3.7).

4.4.2 Dimensional. Each sample unit selected in accordance with 4.3.3.2 shall be measured to determine compliance with the dimensional requirements (see 3.5) of this specification.

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4.5 Examination of preparation for shipment. Examination of the packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.6 Tests.

4.6.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed in accordance with method 111 or method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1. In case of dispute, the analysis by method 111 shall be the basis for acceptance.

4.6.2 Hardness.

4.6.2.1 Forgings for time fuze rings. Hardness readings shall be taken on each sample forging as follows:

At least 3 points at approximately equal distances around the large diameter surface.

At least 3 points at approximately equal distances around the small diameter surface.

At least 3 points at any cross-section indicated in the contract or order.

4.6.2.2 Other forgings. Hardness readings shall be taken on such portions of forgings other than time fuze ring forgings as specified on applicable drawings.

4.6.2.3 Test method. Hardness tests shall be conducted in accordance with ASTM E 18.

4.6.3 Macroscopic examination. The sample selected for macroscopic examination shall be longitudinally sectioned, suitably prepared, etched and visually examined to determine compliance with the requirements of 3.3. Samples for macroscopic examination may be selected from forgings used for the hardness test.

4.6.4 Mercurous nitrate test. The mercurous nitrate test shall be conducted in accordance with ASTM B 154.

4.7 Rejection.

4.7.1 Examination defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.3.3.1 and 4.3.3.2 for that sample size, the entire lot shall be rejected, subject to the provisions of the section on "Disposition of Nonconforming Product" of MIL-STD-105.

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4.7.2 Test failures. A lot shall be rejected for failure to meet any of the test requirements when subjected in accordance with 4.6, subject to the provisions of the section on Rejection and Retests (section 4 of the General Section) of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing. Unless otherwise specified in the contract or order, the material shall be packed in accordance with level C of MIL-C-3993 (see 6.3).

5.2 Marking. In addition to any special marking required in the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brass forgings are intended for use in the manufacture of time fuze parts and other parts.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Dimension and tolerances (see 3.5).
c. When item identification marking is required (see 3.6).
d. Lot size for forgings weighing over 1/2 pound each (see 4.2).
e. Number of tests for forgings weighing over 1/2 pound each (see 4.3.2.2).
f. Location of cross-sectional hardness tests on forgings for time fuze rings (see 4.6.2.1).
g. Location of hardness tests on forgings other than time fuze ring forgings (see 4.6.2.2).
h. Maximum gross weight of containers with contents.
i. Special marking, if required (see 5.2).

6.3 The requirements for item identification marking (see 3.6) and for packing and marking for shipment (see 5.1 and 5.2) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
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